17 n is an integer.

Prove algebraically that the sum of $\frac{1}{2}n(n+1)$ and $\frac{1}{2}(n+1)(n+2)$ is always a square number.

Expand the products before taking one half:

×	n	$+1$
n	n^2	n

×	n	$+2$
n	n^2	$2n$
$+1$	n	$+2$

1 mark
Expand both expressions

Sum = $\frac{1}{2}(n^2 + n + n^2 + 3n + 2)$

$= \frac{1}{2}(2n^2 + 4n + 2) = n^2 + 2n + 1$

×	n	$+1$
n	n^2	n
$+1$	n	$+1$

$= (n + 1)^2$

$n + 1$ is an integer, so its square is always a square number.

1 mark
Identify the square of an integer

(Total for Question 17 is 2 marks)